THE UNIVERSITY OF NOTRE DAME AUSTRALIA

MED 4000
WEEK 24

TUTOR GUIDE

The cases and materials in this guide were written and developed by;

- Professor Joseph Rey – Head of Psychiatry
- Associate Professor Rosa Canalese – Associate Dean Teaching and Learning.

The cases and materials in this guide were reviewed by;

€ Members of the University of Notre Dame clinical year's curriculum committee [CYC]: Last Reviewed July 2010.

RELEVANT PRIOR LEARNING

MED 1000

1.09 – Developmental aspects of adolescence. [L]
1.13 – History and Examination of the adolescent patient. [L]
MED 1000 communicating with adolescents – workshops

MED 2000

Paediatric development workshops

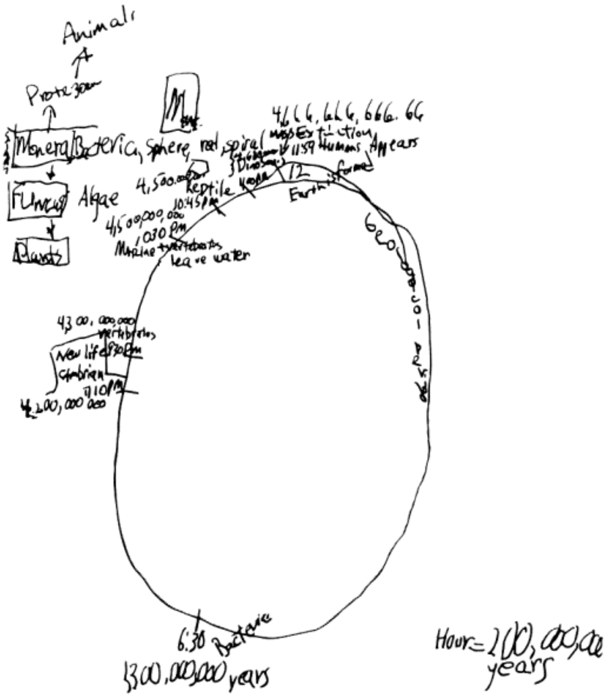
WEEK 24_PSYCHIATRY_PAEDIATRICS		
YR	**WK**	**LEARNING OBJECTIVES**
4	24	• Demonstrate an understanding of the specific issues involved in the assessment of children and adolescents with mental health problems • Outline the main psychiatric disorders common during childhood and adolescence • Know the basic developmental themes of infancy and the pre-school years (0-5) • Be able to describe the disorders that occur during infancy and the pre-school years (0- to 5), their symptoms and basic management • Be able to distinguish between autism and Asperger's disorder and detail themanagement of each condition
4	24	• Know the basic developmental tasks of primary school-age children (6-12) • Demonstrate an understanding of the following childhood anxiety disorders, generalised anxiety disorder, panic disorder, separation anxiety disorder, single and social phobias and obsessive compulsive disorder, by being able to outline their characteristics, clinical presentations and treatment. • Know the therapeutic processes involved in the cognitive behaviour therapy of childhood anxiety disorders. • Describe the differences between school refusal and separation anxiety.
4	24	• Describe the following types and manifestations of childhood externalising disorders, Attention deficit hyperactivity disorder [ADHD], oppostional defiant disorder and conduct disorder. • Be able to diagnose ADHD • Know the pharmacological and non- pharmacological treatments of ADHD as well and be able to explain the indications, contraindications and potential side effects. • Summarise the issues and controversies surrounding ADHD and its treatment • Know the broad groups of specific learning disorders and their association with ADHD.
4	24	• Know the prevalence and changes in drinking patterns among young people • Be able to identify and manage *acute alcohol intoxication ['alcohol poisoning']*. • Know the key strategies to reduce alcohol consumption in young people. • Know the epidemiology, symptoms, diagnosis, and treatment of oppositional defiant and conduct disorder. • Know how to manage 'out of control' young people presenting to emergency services

CASE ONE

Short case number: 4_24_01
Category: Mental Health and Human Behaviour
Discipline: Psychiatry
Setting: GP practice
Topic: Autism spectrum disorders

CASE

Mrs Wong comes to see you seeking advice about her son Robert, aged 11. He is being increasingly teased and bullied at school because of his unusual behaviour. Robert always had difficulties relating to peers and spends his time alone, mostly pursuing his all-consuming interest in astronomy to the exclusion of everything else (see drawing below). Teachers think that he may have autism.



[a] Drawing illustrates the history of the universe from the moment of its creation (12:00 midnight) through geologic time, e.g., the appearance of bacteria (6:30 a.m.). It illustrates the patient's profound interest (and knowledge) regarding this topic, which tended to be all-encompassing, as well as his less-developed fine motor abilities.

QUESTIONS

1. What are the key issues in the psychiatric assessment of children and adolescents?
2. Outline the common mental disorders found in children and adolescents.
3. To understand the context of Robert's behaviour, discuss briefly the developmental aspects of infancy and pre-school years (0-5)
4. What are the characteristics of common mental disorders found during infancy and the pre-school years?
5. In light of Robert's symptoms, explain the key differences between autism and Asperger's disorder
6. **6.** Following your initial assessment of Robert, you are concerned that he has some features of Autism, what would be your next step?

Suggested reading:

- **Management of Mental Disorders (fourth edition) Volume 2. World Health Organisation, Collaborating Centre for Evidence in Mental Health Policy. Sydney 2004. Pp516-540.**

Other resources:

Autism, Asperger's disorder, developmental disability
http://www.wpanet.org/uploads/Education/Educational_Resources/

ANSWERS

Question 1
What are the key issues in the psychiatric assessment of children and adolescents?

As illustrated in Robert's case, children do not usually seek help themselves, their problems are often defined by another person and therefore clinicians must initially explore what is the problem from that person's perspective. Child and adolescent psychiatry is firmly set in the context of the family and the developmental experience of the child, but also in relation to the network of caregivers and social and cultural expectations. Many symptoms of childhood psychiatric disorders are examples of the range of emotional and behavioural reactions to every day events such as changing school, teasing, parental illness and exam pressures. The first challenge is to determine the pattern of symptoms, their relationship to biological, psychological, social and cultural aspects of development, their impact on daily life and adjustment (for example on progress at school), and their relationship to family function. The second challenge is to determine whether the symptoms fulfil criteria for psychiatric disorder(s) and whether treatment is available or indicated.

It can be difficult for children to describe their symptoms and emotions: they might not easily trust the clinician or feel comfortable divulging distressing, perplexing, embarrassing or socially unacceptable behaviours or thoughts. The use of drawings and play may assist in this process.

Assessment
Assessment of a child or adolescent differs from that of an adult in several respects:
- While many adults seek help on their own behalf, children rarely do so.
- Parents generally are the primary source of information, while the patient provides supplementary details.
- Greater emphasis is placed on a corroborative history from teachers or other professionals who know the child.
- The developmental level of the child has to be taken into account when considering presenting features.
- The effect of the problem on development as well as current function is stressed.
- Assessment requires more time (2-5 hours). Briefer assessment is undertaken in an emergency, but this is followed by more extensive evaluation at a later point.

Clinicians vary in whom they see at the first appointment. We recommend seeing the child with mother and father whenever possible. The venue should be private, large enough to accommodate family members, and provide age appropriate toys and related materials. Allow the first 10-15 minutes for the family to offer a relatively uninterrupted narrative. The experienced clinician attends not only to the content, but also to the interactions between family members. The parents' accounts may not concur, and parents', child's and teachers' reports may also be discrepant. Adults are more likely to report behaviours such as impulsiveness and aggression, and children feelings like sadness and worry. It is common to supplement the history with questionnaires completed by parents or teachers. This permits quantification of symptoms, outcome evaluation (when questionnaires are administered before, during and after treatment) and detection of problems that may have been overlooked.

Assessment goals
The first is to establish the nature and context of the presenting problems, their precipitants, attribution and impact on functioning. The second is to obtain a picture of the child's development in the family context. Of particular relevance are details about early infancy and the quality of attachment between mother and infant, the child's transition from one stage to the next (e.g., preschool to primary school), cognitive and school functioning, peer and family relations, physical development, medical history, and temperamental traits such as responsiveness, irritability and impulsivity. The clinician seeks evidence of developmental vulnerabilities and strengths. The third is to gain an understanding of parents' and overall

family functioning, including the marriage, quality of sibling relationships, and influence of extended family. The final goal is to obtain details about medical and psychiatric disorders in other family members, nuclear and extended.

Assessing children

Assessing the child separately is always undertaken, but is most relevant with internalising symptoms or where sexual or physical abuse is suspected**.** Very young children are best engaged through interactive play, but children from about 8 upward may be interviewed directly. Care is taken to use language appropriate to the child's mental age. Ask children why they have been brought to the clinic. Emotional difficulties and preoccupations may be elicited by asking what makes them happy, worried, angry or sad. Formal and informal 'projective' techniques are often used (e.g., asking children how they would spend three wishes, whom they would take to a deserted island for company, and what animal they would most like to be). Drawings also reveal internal preoccupations and concerns. Structure may be introduced by asking children to draw a person, the family doing something, a dream or imaginary television show. Confidentiality can be assured with respect to specific content, but it is important to inform the child that parents will wish to learn of the clinician's general impressions.

Assessing adolescents

Psychiatric examination of an adolescent resembles that of an adult, including inquiry about symptoms and mental state examination. The clinician usually arranges to see the adolescent alone first, before conducting a conjoint interview with the parents (some older adolescents may not wish to involve parents; generally this should be respected). This conveys the message that one is interested to hear what the adolescent has to say and facilitates confidentiality. Adolescents should be asked if there is anything they would not wish to have discussed with parents, but one should be honest about the limits of confidentiality, since risk to self and others and specific illegal activities cannot be kept secret.

The clinician is realistic in not expecting to necessarily develop a working alliance with an adolescent at the first encounter. Silence is used sparingly since it may be perceived by the adolescent as threatening. Similarly, less direct eye contact than with older patients may facilitate an evolving rapport. Finally, clinicians should be genuine: neither trying too hard to identify with adolescent culture nor behaving in authoritarian fashion.

Investigations

The need for tests is dictated by the nature of the problems or medication. Depression may point to blood and endocrinal screening, psychotic symptoms to a CT or MRI scan. Prescription of lithium requires thyroid function testing and other baseline investigations. Psychometric and educational achievement tests may be indicated. Indeed, complex cases may require a child neuropsychologist's expertise. Some children need a speech pathologist's assessment of language problems, or occupational therapist's of poor coordination.

Question 2

Outline the common mental disorders found in children and adolescents

Labelling children with a diagnosis when individual development and family and social circumstances vary so much is problematic. Moreover, difficulties are often situational or improve with maturation (e.g., bedwetting). Little is known about the causes of most disorders, in contrast to risk factors and associated circumstances, like socioeconomic status. Finally, some diagnoses have limited implications for treatment since their effectiveness (e.g., for conduct disorder) is not yet established. Nevertheless, classification is essential for communication, to research aetiology and natural history, and to clarify response to treatment. Optimal management takes into account both the characteristics a child shares with others (diagnosis) and developmental, family and social circumstances.

In making a diagnosis it is important to keep in mind, first, that problems are seen in the context of normal development, i.e., what is common for 1-year-olds can be maladaptive for 7-year-olds (e.g., nocturnal enuresis, separation anxiety). Secondly, disorders are often about a failure to achieve developmental milestones. Finally, symptoms may be diagnosable because of their intensity or persistence; thus temper tantrums are common in pre-schoolers, but deemed problematic only if they occur frequently, are of long duration and lead to functional impairment.

With the exception of certain developmental disorders, discrete conditions are difficult to identify in infants. During preschool and primary school years symptoms and disorders become clearer with age.

Overall, conditions are divided into broad groups:

- Emotional or internalising (e.g. anxiety and depression)
- Disruptive or externalising (e.g. ADHD, conduct disorder)
- Elimination (e.g. functional enuresis, encopresis)
- Developmental (e.g. learning disorders, mental retardation, autism)
- So-called ‘adult’ disorders that can start in childhood or adolescence (e.g. anorexia nervosa, substance abuse, schizophrenia, bipolar).

Question 3
To understand the context of Robert’s behaviour, discuss briefly the developmental aspects of infancy and pre-school years (0-5)

Social functioning evolves rapidly during the first three years, paralleling language development. At 8 weeks babies respond to any face with smiling and eye contact. By 6 months this response is specific to the main caregivers, usually mothers and other family members, who assume particular importance for older babies and mobile toddlers. At times of emotional distress, hunger, illness or separation, it is to these persons (attachment figures) that children look for comfort. Attachment figures also facilitate children’s exploration of the environment. Disruption of these relationships (e.g. through separation) can have a profound emotional effect, possibly resulting in subsequent detachment (lack of response) or anxious attachment (exaggerated distress). Recovery can take weeks or months. Together with these interpersonal developments, the child’s self-concept begins to evolve, both laying the ground for future social ties.

Infants differ considerably in terms of temperament — some are regular in their habits (feeding, sleeping), placid and easy to soothe (‘easy’ temperament), others intense, poor sleepers or feeders and difficult to comfort (‘difficult’ temperament). Such qualities have long-term implications. For instance, ‘difficult’ infants make much greater demands on caregivers and their needs may not be met if a parent, especially mother, is depressed or anxious.

Disorders in the first five years usually reflect a disturbed infant-caregiver relationship, developmental (e.g. low intelligence) or organic (e.g. brain dysfunction) problems. Disordered sleep or feeding may be a function of a biological problem like poorly coordinated sucking, but can be a response to environmental factors. For instance, a poor fit between the infant’s and caregiver’s personalities can generate a spiralling dysfunctional interaction, increasing both infant and maternal distress.

Careful assessment of the infant’s health, developmental abilities and temperament, parent-infant interaction and parents’ expectations and mental health are all relevant in determining treatment: training parents in behaviour management skills (e.g. not rewarding behaviours which maintain disturbance); teaching helpful behaviour in small steps (e.g. putting a child to bed slightly earlier each evening over a period of weeks); creating normal developmental expectations; and treating a parent who has a mental health problem in his or her own right.

Question 4

What are the characteristics of common mental disorders found during infancy and the pre-school years?

Disorders of infancy and early childhood (0-5 years)	
Disorder	**Characteristics**
Failure to thrive	The infant presents with delayed motor and language development, indiscriminate clinging and periods of withdrawal. This is in response to grossly distorted care or neglect which also leads to a failure to gain weight. Management usually requires in-patient care.
Elimination disorders	**Functional enuresis** Children who do not have a physical disorder like urinary infection, epilepsy or diabetes, and who wet their bed or clothes, day or night, at least twice weekly for one month after the age of 5 are diagnosed as having functional enuresis. Primary enuresis: where the child has never been dry for a long period, is not generally associated with emotional or behavioural problems. A family history of wetting is common. Secondary enuresis is wetting that follows at least 12 months of urinary control. Recurrence is often linked to family conflict or stressful events (e.g. abuse, parental separation) and accompanied by behavioural and emotional difficulties. Complete psychiatric assessment, physical examination and urine testing are required. Behavioural treatment using a bell and pad alarm is most effective. Parents help the child to use the toilet when the alarm goes off and they or the child reset it after changing the bed. Treatment continues at least until one month of dry nights. Desmopressin (a vasopressin analogue), the first medication of choice, is effective in the short term but associated with a high relapse rate and side-effects. It is particularly useful to control symptoms when children sleep away from home.
	Functional encopresis It is typified by the repeated passage of faeces into inappropriate places like clothing in the absence of an organic problem. Encopresis is usually involuntary but can be intentional; it occurs in 1% of 5-year-olds, more frequently in boys. It is often due to 'overflow' incontinence resulting from chronic constipation, which may be associated with a diet low in fibre and high in sugar, as well as anal pain. Once organic problems are excluded, treatment focuses on achieving regular defecation. This involves rewards for sitting on the toilet at specified times, a high-fibre diet (associated with judicious use of laxatives if needed) and parental guidance to guard against negative reactions towards soiling. A short admission may be needed to empty a severely blocked bowel in severe cases. Comorbid conditions need to be treated actively.
Autism spectrum disorders	**Autism (see question 5)**
	Asperger's disorder (see question 5)

Disorders of infancy and early childhood (0-5 years)	
Developmental disability (formerly mental retardation)	Significantly impaired intellectual (IQ below 75) and adaptive functioning that leads to psychosocial problems. It is classified according to severity as **mild** (55-75), **moderate** (40-55), **severe** (25-40) and **profound** (below 25). Intelligence testing is a pre-requisite for diagnosis. Moderate and severe forms are identified early because developmental milestones are markedly delayed. Milder forms declare themselves during primary school as a result of academic difficulties. About one in a hundred people are developmentally delayed, most commonly of the milder type. Aetiology is heterogeneous. Because of improved antenatal care, injury, infections and toxins have become less prevalent causes while genetic factors have become more prominent (eg. Fragile X Syndrome: 5%; Down's Syndrome: 15%-20%, the most common). No specific aetiology can be found in up to 40%, but X-linked abnormalities are the likely cause. Environmental influences (eg. malnutrition, emotional and social deprivation experienced, for example, in poorly run orphanages) can also cause or aggravate mental retardation. About one in four developmentally delayed adults and children display intense, frequent or prolonged episodes of aggressive or self-harming behaviour ("challenging behaviour") which disrupts access to community services, theirs and carer's safety. Challenging behaviour can be a reaction to an inadequate environment and often lessens when parenting or institutional practices change by improving behavioural management. Drug treatment is often used in these cases, so much so that about 20% are prescribed antipsychotics. These drugs are useful in the acute management of aggression but ineffective or harmful in the long term.

Question 5
In light of Robert's symptoms, explain the key differences between autism and Asperger's disorder

Children with autism show abnormal social interactions (eg. inability to appreciate the feelings of others, lack of empathy and imaginative play), problems in verbal and non-verbal communication (eg. rudimentary or difficult to understand speech, repetition of words and sentences), and narrowing of activities and interests (eg. rocking, intolerance of change, repetitive rituals and mannerisms, fascination with moving objects, hand flapping or twisting). Onset is before the age of three. Lack of responsiveness is often the first symptom noticed by parents, so much so they often think their child may be deaf. Mental retardation occurs in most cases although a small proportion has normal cognitive abilities and are referred to as 'high functioning.' Some display unusual abilities (e.g., in music, drawing or calculation—'savants'). Neuropsychological assessment shows a scatter of scores on different subtests. The burden on families is high.

Autism is an uncommon condition (1 per 1000) across all the socioeconomic levels; it affects males three times as often as females. Psychosocial factors or abnormal parenting are not the cause, rather it is a neuropsychiatric condition of unknown aetiology. Illnesses like genetic abnormalities (eg. Fragile X Syndrome), rubella encephalopathy, auto-immune disorders and epilepsy occur frequently.

Symptoms are life-long; those with higher intelligence or language skills are more likely to improve (eg. learn to care for themselves, travel independently, communicate). Adolescence is a difficult period. A small number manage to live and work independently as adults, one third achieve partial independence (eg. sheltered work, group-home living). A typical autistic adult is well played by Dustin Hoffman in the film

Rainman.

Early diagnosis is important because outcome improves with multimodal and intensive treatment: education, behavioural strategies and social skills training for the child, information, support and behaviour management training for the parents. Drug treatment of symptoms such as aggression and anxiety, and specific management of comorbid conditions like depression, ADHD, tic disorder and epilepsy also affect outcome as well as reducing the family burden.

Asperger's disorder

This condition shares with autism social and stereotyped behavioural features but not delayed and deviant language development—this being the key difference between Asperger's and autism. Prominent symptoms are inability to use non-verbal behaviour to modulate social interaction (eg. lack of eye contact, facial expression, understanding of social norms and empathy) and a limited repertoire of interests and activities (eg. inflexible adherence to routines, preoccupation with parts of objects, infatuation with concrete topics like train time-tables or street maps). Because they have normal IQ and can communicate, symptoms are recognised later than in the case of autism, at school or during adolescence. However, psychosocial impairment and family burden can be just as prominent. The frequent comorbidity (eg. Tourette's Disorder - recurrent motor and vocal tics - anxiety, depression) makes diagnosis difficult and complicates the presentation. Some repetitive behaviours and obsessive preoccupations may resemble obsessive compulsive disorder. Diagnosis of Asperger's disorder has increased considerably in recent years. The reasons for this, whether improved detection, increased prevalence or widening of the diagnostic criteria, are not known.

Characteristics persist into adulthood with some developing idiosyncratic, mainly schizoid, personality traits. Most individuals achieve some adaptation, are able to function independently and even be successful professionally, but not socially. They have an increased risk of psychotic episodes.

Treatment

There is no specific treatment for autism. There is some evidence that the earlier behavioural, educational, speech, and occupational therapy is begun, the better the long-term outcome. This is often an intensive, demanding, and long-term commitment. Medication, particularly atypical antipsychotics (e.g., risperidone) have been shown to be effective in managing 'challenging behaviour' such as aggression and temper outbursts. There are many treatments for Asperger's disorder, particularly focussing on developing social skills. Evidence for their effectiveness is limited. In all cases, comorbidities, which are frequent, should also be treated.

<u>Question 6</u>

<u>Following your initial assessment of Robert, you are concerned that he has some features of Autism, what would be your next step?</u>

Autism is a severe, life-long condition for which, as noted, there are limited treatments, which have little impact on the symptoms of the disorder. Therefore, one should be circumspect about making such a diagnosis without a thorough evaluation (e.g., psychometric testing) and a specialist's opinion.

Once the diagnosis is confirmed, you may then discuss with the family the treatment options, including access to government-funded support. Since a diagnosis of autism or Asperger's disorder allows access to government-funded support, there has been concern that these conditions, particularly Asperger's disorder, are being overdiagnosed.

The Commonwealth Government has committed funds for the four years up to June 2012 to deliver the 'Helping Children with Autism' package. The package will help address the need for support and services for children with autism spectrum disorders. The package includes:

- Autism Advisors
- Funding for early intervention services: The funding supports the delivery of multidisciplinary evidence based early intervention to facilitate improved cognitive, emotional and social development prior to a child starting school.
- Specialised playgroups
- Family workshops

Following diagnosis, families contact an Autism Advisor who will provide information regarding eligibility, available funding and early intervention and other support services. The Department of Health and Ageing has made new Medicare items available for children aged under 13 years (for diagnosis and treatment planning). If the family lives in an outer regional or remote area or has two or more children with a diagnosed disability they may be eligible for a $2,000 Outer Regional, Remote and Access Support Payment.

Autism Spectrum Disorder Website

The Raising Children Network (RCN) website (http://raisingchildren.net.au/articles/hcwa_funding_partner.html) provides information, online resources and interactive functions to support parents, families, carers and professionals. The RCN is an Australian Government funded site providing information to parents of children from birth to school age, including:

- Evidence based information about ASDs and early intervention approaches
- Information on diagnosis and assessment
- A guide to navigating the service system
- Film clips
- Parent forums.

CASE TWO

Short case number: 4_24_02
Category: Mental Health and Human Behaviour
Discipline: Psychiatry
Setting: GP practice
Topic: Separation anxiety disorder

CASE

Mrs White comes to your GP surgery with her 10 year old daughter Fleur. Fleur had vomited that morning when she was due to leave for school. She was complaining of stomach ache and headache. Physical examination shows no abnormalities. You learned that Fleur had initially been at home with a 'cold' for a few days but has now missed 2 weeks of school. Fleur feels sick in the morning, and looks pale and unwell. "I can't send her to school while she is so sick… I am so worried that it is something serious"

QUESTIONS

1. Your assessment of Fleur does not suggest any sinister cause for her illness; how would you respond to her mother's concerns?
2. Summarise the key developmental issues of the primary school years (6-12) that may provide a context for Fleur's behaviour.
3. As you further assess Fleur, there are features that are suggestive of an anxiety disorder. Describe in a table the anxiety disorders in children and adolescents, their characteristics, clinical presentations and their key treatment?
4. What are the key differences between separation anxiety and school refusal?
5. Fleur is subsequently diagnosed with generalised anxiety disorder; briefly outline the use of cognitive behavioural therapy in the management of children with anxiety disorders.

Suggested reading:

- **Management of Mental Disorders (fourth edition) Volume 2. World Health Organisation, Collaborating Centre for Evidence in Mental Health Policy. Sydney 2004. Pp516-540.**

Other resources:

- **How to treat anxiety disorders in children and adolescents. Australian Dr 5 October 2007.**

http://www.australiandoctor.com.au//htt/pdf/AD_031_038_OCT05_07.pdf

ANSWERS

Question 1
Your assessment of Fleur does not suggest any sinister cause for her illness; how would you respond to her mother's concerns?

Parents' attitudes and response are critical to the management of most childhood mental disorders, particularly anxiety disorders. Parental responses, if inappropriate can reinforce the child's anxieties becoming more entrenched and chronic (e/g/, parental anxiety often feeds the child's anxiety).

Parents often become co-therapists (i.e., implementing at home interventions discussed in the consultation— homework). Therefore, it is important from the start to engage (rather than alienate) the parents and educate them on the possible diagnoses and the rationale for treatment. In the case of Fleur, this may entail reassuring Mrs White that Fleur's symptoms are not serious, and do not seem the manifestation of a physical illness. One may also enquire wether Fleur's complaints lessen once she realises she does not have to go to school (e.g., is she OK the rest of the day?). The mother needs to be told that if Fleur does not go to school, the longer she remains away from school the harder it will be her return.

Question 2
Summarise the key developmental issues of the primary school years (6-12) that may provide a context for Fleur's behaviour.

During the primary school years, children gradually move away from focusing on themselves and on the parent-child dyad and develop increasingly complex relationships with others. Social skills acquired in the preschool years equip them to meet the demands of school and peers. Anxiety about separation is a normal part of development in infants and toddlers, particularly around the age of 18 months to four years, and as such is not considered an anxiety disorder. By age 5 almost all children tolerate separation from attachment figures, allowing them to attend school. During the primary school period, children grow in size, physical strength, agility, social skills and cognitive aptitudes. By early adolescence, children think in abstract terms. A smooth transition to school and other social settings is facilitated by a caring and supportive family during infancy; good adjustment to primary school aids the passage to adolescence.

Learning difficulties, deficits in social skills, and emotional and behavioural problems can disrupt the achievement of age-specific goals which in turn have longer term implications. Problems tend to lead to a mismatch between abilities and scholastic demands. Angry outbursts and aggression are common in infancy and early childhood, temper tantrums peaking between 2 and 4. Thereafter, physical aggression declines but its verbal counterpart becomes more prominent. Responses to aggression also change across childhood: preschoolers do not react consistently but most school age children do. By about 8, most children distinguish intentional from accidental events and respond accordingly. With reduced aggression comes 'prosocial' behaviour: taking turns, sharing with others, cooperating on a task and helping distressed peers. This leads to the growth of empathy and capacity to resolve conflict verbally. In some circumstances, aggressive behaviour continues, becoming more elaborate in those with conduct disorder. Parents' responses to early aggressive behaviour influence the pattern. An inconsistent, critical or punitive attitude is likely to result in persistent aggressive behaviour through to adolescence, even adulthood.

Question 3
As you further assess Fleur, there are features that are suggestive of an anxiety disorder. Describe in a table the anxiety disorders in children and adolescents, their characteristics, clinical presentations and their key treatment?

Disorder	Description	Common presentations	Evidence-based treatments
Generalised anxiety disorder	Persistent and excessive worry about normal activities, particularly about adequate performance (e.g., in tests), which results in tiredness, irritability, sleep problems, difficulties concentrating, etc.	• Excessive worry about exams or other activities • Feeling tense and on edge all the time • Difficulty concentrating: "mind goes blank"	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training
Panic disorder	Repeated panic attacks, apprehension about their recurrence and worry about their implications (e.g., "Am I going crazy?")	• Acute, short-lived episodes during which children feel they may have a heart attack, go mad or loose control • Sudden episodes of palpitations, butterflies in the stomach and other anxiety-related physical symptoms • Patients request reassurance that they are not ill	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training • Breathing techniques
Separation anxiety	Persistent and excessive worry about being away from home or from the parent/s that prevent children from leading a normal life (e.g., attending school, sleeping over at friends' places).	• Refusal to go to school often • Complaints of minor physical problems (headaches, nausea etc.) resulting in frequent school absences • Reluctance to go out of the house alone or sleep over at friends' places • Worry that something bad may happen to parents	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training • Exposure to separation Medication: • SSRIs
Single phobias	Irrational fear and avoidance of specific animals (eg, insects), objects (eg, needles, blood) or situations (eg, heights, flying)	Distress and avoidance of animals, objects or situations	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training • Exposure to the feared object or situation
Social phobia	A marked and persistent fear of social situations, often leading to avoidance, because of worries of being humiliated or embarrassed	• Refusal to go to school because of embarrassment about physical education classes, having to read or perform in front of the class etc. • Avoidance of parties, sports or other social situations • Excessive shyness or self-consciousness	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training • Exposure to social situations
Obsessive–compulsive disorder (OCD)	Obsessions, compulsions or both that cause marked distress, are time consuming (eg, take more than one hour a day) or interfere significantly with the functioning of the young person	• Repeating certain acts (checking things, cleaning, washing) again and again • Anger or distress when not allowed to perform some actions the way they want • Intense preoccupation with tidiness, cleanliness, germs, etc. • Need to repeat things until they are "right'"	Cognitive behaviour therapy: • Cognitive restructuring • Relaxation training • Response prevention Medication: • SSRIs

Disorder	Description	Common presentations	Evidence-based treatments
		• Unusual slowness and perfectionism • Skin lesions, raw skin (because of OCD-induced washing rituals)	

Question 4
What are the differences between separation anxiety and school refusal?

Separation anxiety is the only anxiety disorder characteristic of childhood and adolescence. Separation concerns should be considered pathological when they occur after the age of five years, and are developmentally excessive, resulting in distress and impairment in day-to-day function. Separation anxiety disorder is the most common anxiety disorder in children, with rates of about 4%, primarily in children under 12. It affects both sexes equally, with the most common age of onset between six and eight years. Because young people with this disorder frequently refuse to go to school, it is also called *school refusal* and *school phobia*. School refusal is a general, descriptive term. The term *school phobia* is appropriate for children who refuse to go to school because something in the school is frightening them. Often this is fear of being bullied or victimised, but it can be for other reasons. Only some children who refuse to go to school have separation anxiety. A few may be depressed and lack the energy and motivation to attend school, some may be afraid of school for one reason or another (having no friends, feeling picked on by teachers, being embarrassed about exercising in front of peers in physical education classes), others may just dislike school and school work and prefer to be at home.

Young people with separation anxiety become very worried when they are away from home or from their parents. They can't stand the separation. Typically this results in their refusing to go to school. However, what bothers most of these children is not school itself – usually they are good, well-liked students – but the fact that attending school means being away from parents and home. What they want to avoid is separation from parents or the house.

This worry is so intense that these children can become physically ill (headaches, vomiting) in the morning when they are due to leave for school. Mondays and the days following school holidays are the worst. On other occasions they need to be taken home from school because they feel sick. Often they are reluctant to sleep over at friends' places, or feel uncomfortable about travelling by themselves or going on errands. When at school, they worry about something happening to the parents (a car accident, rape, burglary, heart attack) and have vivid fantasies about it. Occasionally, they can't cope with parents going out or, if the parents do go, need to know where, the phone number where they can be contacted, and so on. It is not unusual for these teenagers to have difficulties going off to sleep or to need the company of a parent when they go to bed.

The onset of this condition is often triggered by worrying incidents, such as an illness or death in the family, a separation or divorce, a change of school or a minor illness that keeps the teenager at home for a few days, but it can emerge out of the blue. Typically, there are times when the child appears not to be worried by separation and periods when it becomes worse.

Separation anxiety disorder is a common condition that afflicts as many as 4 per cent of all children and teenagers. Its frequency peaks during early adolescence and then decreases. Some adults still experience severe discomfort when separated from partner or children. Minor worries about separation are frequent among adults (e.g. if your child is due home at ten at night and does not arrive, it is quite normal to worry about whether something might have happened).

Question 5

Fleur is subsequently diagnosed with generalised anxiety disorder; briefly outline the use of cognitive behavioural therapy in the management of children with anxiety disorders.

The better researched are the psychological treatments that are subsumed under the cognitive behaviour banner. At least 10 hours of face-to-face treatment as well as homework is usually required.

After a thorough assessment and identification of the problems, the first stage of treatment is explaining the nature of the disorder to the young persons and their families and obtaining their co-operation. Explaining to young people that these problems are not normal and require treatment is difficult. Skilled clinicians use drawings, pictures and other aids to explain the condition and its symptoms to children. This is a delicate phase but essential for future success. For example, treatment of separation anxiety will require teenagers to attend school, experience the separation from parents and realise that nothing will happen. Children who wash their hands repeatedly because of contamination fears will need to tolerate having their hands dirty and resist the urge to wash them for increasingly longer periods of time. Lack of cooperation by the young person or the family will make treatment very difficult.

Parental involvement

Families of children with anxiety disorders often become involved in their children's fears or suffer as a result of them. For example, parents or siblings may resent the need for reassurance of anxious children and adolescents or their rituals in the case of obsessive–compulsive disorder (e.g., the bathroom is occupied for long periods and large amounts of hot water or soap are used). The social life of the whole family is often disrupted because of teenagers' delays because of compulsions or refusal to attend school, family or social gatherings.

Family members are often drawn into performing some of the rituals or forced to keep special sets of cutlery, towels or other everyday items. For example, a 13-year-old boy refused to go to sleep every night until his mother recited a series of prayers three times. When she refused, he became distressed, angry and abusive. Parents' work may also be affected if they have to accompany the child to school because of intense distress or if they cannot go to work because the child refuses to attend school. These matters need to be discussed and family members made aware that these behaviours are not just "naughty" but the manifestation of an illness.

Parents need to be closely involved in the treatment, particularly when children are younger. In many instances parents should become co-therapists and help the child carry out tasks that need to be done at home, following the therapist's instructions (homework). For example, the father may be asked to take the child to school in the morning in cases of school refusal despite the child's reluctance to attend.

The goals of treatment, as well as what to do when children resist treatment, need to be explained in detail and agreed upon by all concerned. Parents can also play an important role in monitoring the frequency of specified behaviours (e.g., by keeping a diary).

Cognitive behaviour treatments

These are effective and often result in a rapid improvement of symptoms after a few hours of therapy if the young person is able to follow the treatment through. The basic techniques used are cognitive restructuring, relaxation and exposure. The same processes are used when treating depression although the content varies (e.g., instead of focusing on the fear, the spotlight in on negative self-perceptions: "no one likes me", "I am no good")

Cognitive restructuring

The initial goal is to help the young person understand the links between thoughts, feelings and bodily symptoms. The outcome should be that children recognise their negative thoughts when they occur and the effect they have on their emotions and their body.

This usually entails teaching children the difference between the various emotional states (e.g., fear, anger, sadness, joy). While doing this, young people are trained in quantifying the intensity of their emotions by using graphical aids such as a "fear" or a "worry" thermometer that can be coloured in. This helps them understand there are different feelings and that emotions vary in their intensity.

The next step is to help them understand the relationship between feelings (e.g., fear) and their physical manifestations (e.g., “how does your body feel when you are scared or worried?”). Again, this is often done with drawings of their body where the child can show or draw where and how they feel the various emotions.

Finally, children are helped to identify the thoughts they have when worried or anxious by asking them to imagine situations and the kind of thoughts that come to their mind.

For example, “Imagine that your mother does not arrive at home when you expect her. What would you think?” Anxious children often come up with a negative if not catastrophic thought (e.g., “She has had an accident”).

After this initial response other possible thoughts or explanations are explored. This allows making clear to children that there can be a variety of thoughts in response to a situation. Because they have already learned that different thoughts are associated with different emotions, in this way they recognise their automatic thoughts and self-talk and how these thoughts influence the way they feel.

This process lays the foundation for cognitive restructuring, or the development of realistic thinking as opposed to worried thinking. The exercises required include:

1. Identifying the thought behind the emotion (“What is making you scared?”, “What do you think may happen?”).
2. Looking for evidence to justify that thought.
3. Examining other possible explanations.
4. Evaluating the worrying thoughts on the basis of experience.
5. Examining the consequences of the feared event.

This process should be followed irrespective of the type of anxiety disorder, although the content may vary.

In the case of obsessive–compulsive disorder, it may entail examining the thoughts or feelings triggered by contaminating the hands with dirt, while in a child with school refusal it may include thoughts triggered by going to school or by parents going out of the house for a meal.

Relaxation

Relaxation techniques are an effective adjunct to exposure (see next section) in the treatment of phobias and obsessive–compulsive disorder and a mainstay of treatment for generalised anxiety.

There is a variety of effective techniques to teach young people how to relax; most follow progressive muscle relaxation principles. They require work and persistence before they are mastered.

Exposure

A common element in the treatment of phobias and obsessive–compulsive disorder is exposure to the feared object or situation (in obsessive–compulsive disorder this is called response prevention). Treatment is unlikely to be effective if young people do not actually face what scares them. Tolerating the fear and worry and realising that nothing terrible happens diminishes the anxiety and overcomes the phobia. Consequently, avoidance or compulsions gradually weaken.

Exposure usually takes place as an extension of cognitive restructuring and after relaxation techniques have been learned. Praise and rewards after each success are very important in maintaining the child’s cooperation.

Fears should be faced gradually. This requires the therapist and the child to list the situations that are less and more frightening (e.g., constructing a hierarchy) using a fear thermometer or an aid appropriate for the age of the patient. The child is then encouraged to attempt facing these situations (often after modelling by the therapist or parents), starting with the less worrying tasks, relaxing and moving on to more frightening situations.

It is important the young person remains in the feared situation for long enough, until anxiety dissipates, to realise the things they fear do not eventuate. If patients are left very worried or anxious by the end of the session, treatment could make the situation worse.

Success will be achieved through repetition and practice at home or away from the consulting room (homework). Exposure often requires going to the places where fears are worse (e.g., shopping centres, school) to carry out some sessions.

Parents need to be aware of these issues to support the exercises, particularly when performed at home. Leaving the parents out of the treatment process often causes problems, or parents may end up undermining the treatment.

Breathing

Many anxious young people breathe too quickly and deeply, ie, hyperventilate. Hyperventilation produces vasoconstriction, leading to feelings such as pins and needles in the hands, light-headedness, butterflies in the stomach, dizziness and other symptoms that make the young person feel sick and hence more worried and anxious.

Teaching how to breathe properly to avoid hyperventilating is helpful. Hyperventilation usually occurs during panic attacks and makes them worse.

If children have a panic attack, first reassure them (and parents) that nothing will happen. Second, ask them to breathe slowly and not deeply, with the stomach and not with the chest (eg, breathe in, hold your breath, count mentally to 10, breathe out). Third, if available, breathe in a paper bag until symptoms subside.

CASE THREE

Short case number: 4_24_03
Category: Mental Health and Human Behaviour
Discipline: Psychiatry
Setting: GP practice
Topic: Attention deficit hyperactivity disorder (ADHD)

CASE

John's mother has brought him to the GP at the suggestion of the school counsellor. John is 8 years old, inattentive, overactive, noisy and disruptive in the classroom. This had been a significant problem since he began attending school. In spite of his small size, teachers found him difficult to manage, 'the class clown' and a negative influence on the other students. He had problems with the other pupils and regularly became involved in quarrels. His mother said these problems were not new and that John was also difficult at home: he argued with his brother and sisters, did not do his jobs around the house, and shouted with little reason.

QUESTIONS

1. What would be your next steps in assessing John?
2. You are concerned that John may have ADHD. What are the key features that would support that diagnosis?
3. You explain to John's mother that your preliminary assessment suggests that John may have ADHD. She comments 'every 2nd child seems to be diagnosed with ADHD.' Is it really that common and what causes it? What would you explain to her about the prevalence and aetiology of ADHD?
4. You receive the report from the paediatrician confirming that in her opinion John suffers from ADHD. Summarise the basic issues involved in the management of ADHD
5. What are some of the controversies in the treatment of ADHD?
6. Does ADHD exist in adults?
7. Describe learning disorders and their association with ADHD.

Suggested reading:

- **Management of Mental Disorders (fourth edition) Volume 2. World Health Organisation, Collaborating Centre for Evidence in Mental Health Policy. Sydney 2004. Pp516-540.**

Other resources

- **How to treat ADHD. Australian Dr 28 November 2008**

http://www.australiandoctor.com.au//htt/pdf/AD_025_032_NOV28_08.pdf

- **How to treat speech and language problems in children. Australian Dr 11 December 2009**

http://www.australiandoctor.com.au//htt/pdf/AD_027_034_DEC11_09.pdf

ANSWERS

Question 1
What would be your next steps in assessing John?

As already highlighted, parents' attitudes and response are critical to the management of most childhood mental disorders. Parental responses, if inappropriate can reinforce the child's dysfunctional behaviour. If not properly informed about their child's conditions and the rationale for treatment, they may become uncooperative and make treatment more difficult. Alternatively, they may seek information from the Internet — much of it misleading or poorly digested. Therefore, it is important from the start to educate parents on the potential diagnoses and the rationale for treatment. In the case of John, this may entail:

1. Conducting a brief medical history and examination to ascertain whether there are obvious physical problems
2. Reassuring John's Mother that John's problems are common in boys of that age
3. That problems may have a variety of causes, such as learning difficulties (e.g., reading), a poor 'fit' between John and his teacher/class/school, problems with attention and hyperactivity, or even depression, difficulties with peers or being bullied.
4. A comprehensive history is, of course, required: Does John show the same type of problems at home? Is there family conflict? How impaired is he by his symptoms? Are there symptoms of other conditions? etc.
5. To clarify the nature of John's problems, further investigations will be needed (e.g., IQ testing, achievement tests to elucidate whether there are specific learning problems [see Question 6 below])
6. It will be necessary to contact John's teacher to obtain a first-hand account of his behaviour at school. The teacher will also need to complete some questionnaires to examine the type and severity of John's problems; mother's written permission will be required for that
7. These investigations should probably be conducted by a specialist on the field such as a behavioural paediatrician, child psychiatrist or an experienced child psychologist. Therefore, options for referral will need to be discussed.

Question 2
You are concerned that John may have ADHD. What are the key features that would support that diagnosis?

Children whose conduct is not socially acceptable and who cause disruption or distress to others (e.g. they might be restless, impulsive, disobedient, aggressive, defiant, lie or steal) are said to have 'externalising' or 'behaviour disorders.' These disorders include a variety of problems which may be limited to the child's home and family or which may affect the school or the wider community. Because of the bother these young people cause to the people around them, these problems rarely go undetected and these adolescents are quickly identified as troubled. They are mostly boys who have difficulty taking responsibility for the consequences of their actions and blame others for their problems. While most children can learn that something they have done is wrong after being told so or chastised. The 'externalisers' by contrast do not learn, even after going through such a process many times. Externalising or behavioural disorders include:

- ADHD
- Oppositional defiant disorder
- Conduct disorder

‘Externalising’ contrast with ‘internalising’ or ‘emotional’ disorders, in which children internalise problems, mostly show anxiety or depressive symptoms, and are more common in girls.

ADHD

Children with ADHD are distractible, disorganised, cannot attend to instructions and have difficulty establishing routines. Fidgetiness and excessive running about or climbing are typical of primary school-age, but less so in adolescence. Symptoms occur in more than one setting, usually at home and school, and lead to impairment.

Inattention

Six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent with developmental level and that impact directly on social and academic/occupational activities.

(a) Often ***fails to give close attention to details*** or makes careless mistakes in schoolwork, at work, or during other activities (for example, overlooks or misses details, work is inaccurate).

(b) Often has ***difficulty sustaining attention*** in tasks or play activities (for example, has difficulty remaining focused during lectures, conversations, or reading lengthy writings).

(c) Often ***does not seem to listen*** when spoken to directly (mind seems elsewhere, even in the absence of any obvious distraction).

(d) Frequently ***does not follow through*** on instructions (starts tasks but quickly loses focus and is easily sidetracked, fails to finish schoolwork, household chores, or tasks in the workplace).

(e) Often has ***difficulty organizing tasks*** and activities. (Has difficulty managing sequential tasks and keeping materials and belongings in order. Work is messy and disorganized. Has poor time management and tends to fail to meet deadlines.)

(f) Characteristically avoids, seems to dislike, and is ***reluctant to engage in tasks that require sustained mental effort*** (such as schoolwork or homework or, for older adolescents and adults, preparing reports, completing forms, or reviewing lengthy papers).

(g) Frequently ***loses objects*** necessary for tasks or activities (e.g., school assignments, pencils, books, tools, wallets, keys, paperwork, eyeglasses, or mobile telephones).

(h) Is often ***easily distracted*** by extraneous stimuli. (for older adolescents and adults may include unrelated thoughts.).

(i) Is often ***forgetful*** in daily activities, chores, and running errands (for older adolescents and adults, returning calls, paying bills, and keeping appointments).

Hyperactivity and Impulsivity: Six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent with developmental level and that impact directly on social and academic/occupational activities.

(a) Often ***fidgets*** or taps hands or feet or squirms in seat.

(b) Is often ***restless*** during activities when others are seated (may leave his or her place in the classroom, office or other workplace, or in other situations that require remaining seated).

(c) Often ***runs about*** or climbs on furniture and moves excessively in inappropriate situations. In adolescents or adults, may be limited to feeling restless or confined.

(d) Is often ***excessively loud*** or noisy during play, leisure, or social activities.

(e) Is often ***“on the go,”*** acting as if “driven by a motor.” Is uncomfortable being still for an extended time, as in restaurants, meetings, etc. Seen by others as being restless and difficult to keep up with.

(f) Often ***talks excessively***.

(g) Often ***blurts out an answer*** before a question has been completed. Older adolescents or adults may complete people’s sentences and “jump the gun” in conversations.

(h) Has ***difficulty waiting his or her turn*** or waiting in line.

(i) Often ***interrupts or intrudes*** on others (frequently butts into conversations, games, or activities; may start using other people's things without asking or receiving permission, adolescents or adults may intrude into or take over what others are doing).

(j) Tends to ***act without thinking***, such as starting tasks without adequate preparation or avoiding reading or listening to instructions. May speak out without considering consequences or make important decisions on the spur of the moment, such as impulsively buying items, suddenly quitting a job, or breaking up with a friend.

(k) Is often ***impatient***, as shown by feeling restless when waiting for others and wanting to move faster than others, wanting people to get to the point, speeding while driving, and cutting into traffic to go faster than others.

(l) Is ***uncomfortable doing things slowly and systematically*** and often rushes through activities or tasks.

(m) Finds it ***difficult to resist temptations or opportunities***, even if it means taking risks (A child may grab toys off a store shelf or play with dangerous objects; adults may commit to a relationship after only a brief acquaintance or take a job or enter into a business arrangement without doing due diligence).

Diagnosis is made if symptoms impact directly on social and academic/occupational activities, and are apparent in ***two or more settings*** (e.g., at home, school or work, with friends or relatives, or in other activities). There must be clear evidence that symptoms interfere with or reduce the quality of social, academic, or occupational functioning. The symptoms do not occur exclusively during the course of schizophrenia or another psychotic disorder and are not better accounted for by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, or a personality disorder).

Subtypes of ADHD

- **Combined:** if meets criteria for both inattention and hyperactivity-impulsivity.
- **Predominately Inattentive:** if meets criteria for inattention but not hyperactivity-impulsivity.
- **Predominately Hyperactive/Impulsive:** if meets criteria for hyperactivity-impulsivity but not for inattention.

Question 3

You explain to John's mother that your preliminary assessment suggests that John may have ADHD. She comments 'every 2nd child seems to be diagnosed with ADHD.' Is it really that common and what causes it? What would you explain to her about the prevalence and aetiology of ADHD?

Prevalence

Prevalence estimates of ADHD vary between 4% and 10% according to study, gender and age. Referrals peak between 8 and 11. Boys are affected three times more than girls. Comorbid conditions such as conduct disorder, specific learning disorders, depression and anxiety are common. Clinical features can manifest at preschool but a diagnosis made in very young children is unreliable.

Aetiology

ADHD probably represents a final common neurodevelopmental pathway arising from multiple causal factors, including genetic. Functional neuroimaging shows reduced frontal lobe activity, although this is found in other psychiatric disorders. Food additives were considered a factor during the 1970s but these claims have been discredited. Exposure to critical comments from parents is often noted but whether this is cause or consequence remains unclear. An association between early attachment problems and ADHD is probable.

Question 4

You receive the report from the paediatrician confirming that in her opinion John suffers from ADHD. Summarise the basic issues involved in the management of ADHD

Assessment requires a careful history from parents and other sources such as teachers. Behaviour rating scales for parents and teachers help to quantify symptoms and allow comparisons with similar-aged children. Hearing, visual and neurological problems are excluded. IQ and reading assessment are indicated if intellectual impairment or learning disorders are suspected.

Management is multimodal. The clinician educates parents and child about the nature of the condition, consults with other professionals involved, including teachers. Behavioural counselling helps the parents to increase behavioural compliance but core features (inattention, impulsivity, hyperactivity) respond less well. Techniques include praise and reward for appropriate behaviour, clear instructions (e.g., one specific request at a time) and consistency (e.g., both parents adhere to a simple set of rules).

The school is involved in assessment and informed of diagnostic conclusions and management. Classroom strategies include: presenting work in modules under 10 minutes to avoid overtaxing capacity to attend; interspersing classroom learning with brief periods of exercise; permitting movement provided it does not disrupt peers; using cues to alert students they have strayed from the task; seating them close to the teacher or students who concentrate well; and articulating expectations of behaviour.

Medication

CNS stimulants (methylphenidate and amphetamines) have a pivotal role in reducing inattention, impulsivity and hyperactivity where these interfere with learning, social interaction and family life but should be used in children with moderate to severe conditions. Stimulants are available in formulations that remain effective for eight or more hours (slow release, skin patch), saving the child having to take doses at school (both methylphenidate and amphetamines have a short half-life). The putative mechanism of action is selective stimulation of inhibitory brain pathways. Efficacy and safety in the short term is well established and is likely in the longer term (more than 12 months).

Prescription of stimulants (Schedule 8 drugs) requires a special authority only given to some medical practitioners, usually paediatricians or child psychiatrists.

Common side-effects are loss of appetite, weight loss, sleeplessness and growth retardation. The possibility of sudden death has been raised though evidence is inconclusive. Contraindications include psychotic symptoms, severe tics and other movement disorders, arrhythmias and structural cardiac abnormalities. Weight, height, blood pressure and pulse rate are monitored. Although risk of dependency at therapeutic doses is low, the potential for abuse makes these medications inadvisable when child or family have a history of substance abuse.

In patients where stimulants are ineffective or poorly tolerated, atomoxetine, which is a noradrenergic reuptake inhibitor, may be used as an alternative. It differs from stimulants in having a slower onset of action, with effects persisting for up to 24 hours (compared with 3 hours for stimulants).

Symptoms may moderate sufficiently by late adolescence to warrant medication being stopped. Changes in treatment at critical transition points (e.g. entry into secondary school) are best avoided.

Psychological treatments

Psychological treatments include education and support for parents and cognitive-behaviour therapy (CBT) for the child. CBT aims at helping the child to focus on the task at hand, improving motivation and discouraging off-task activities, i.e., by rewarding on-task behaviours. Clinicians, parents and teachers need to be aware of these children's limitations and set up tasks they can do. Behaviour modification approaches need to be carried out both at home and at school.

These children also benefit from techniques that help them improve self-control and reduce impulsiveness (such as 'stop, think, do': stop what you are doing, think about what you should do now, do it). Because of their difficulties relating to peers ADHD sufferers often require training in social skills.

Psychoeducation, family therapy and parent management training aim to inform the family about the condition and its management and to provide support. An important issue for parents to understand is that their children have an impairment that prevents them from behaving like other children. They are not just 'bad,' although they can do bad things. Telling them they are naughty or punishing them for their disability is not helpful. Parents are also taught ways of managing their children that do not reinforce unwanted behaviour. Programs such as Triple P (Positive Parenting Program) are very useful in this regard and have been shown to be effective particularly in younger children and in those with mild or moderately severe forms of the illness.

Question 5
What are some of the controversies in the treatment of ADHD?

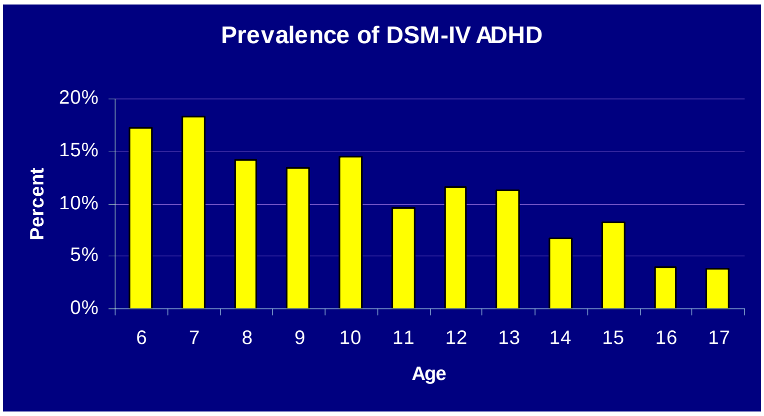
There are still a number of controversies surrounding the treatment of ADHD, highlighted by frequent articles in the mass media. One of them is whether skills are as good as pills for ADHD. In 1992 the US National Institute of Mental Health funded the Multimodal Treatment Study of Children with ADHD (MTA) to examine the long term effects of routine community management *versus* carefully delivered treatments. Children with ADHD, combined type (i.e., showing symptoms of inattention, impulsivity and hyperactivity) were randomly allocated to medication, psychosocial treatment, a combination of both, and routine clinical care. The medication group received an enhanced methylphenidate management in which optimal dose was blindly titrated to achieve maximum benefit. The psychosocial group received a variety of interventions that comprised parent training, a summer treatment camp, and classroom management. Those in the routine care group were given information about services and left to their own devices. 579 children with a mean age of 8.5 years were randomized to the four groups, about 145 in each. The authors concluded that after 14 months the "carefully crafted medication management was superior to the behavioral treatment and to routine clinical care that included medication." The results were influential in treatment guidelines and clinical practice.

Participants were naturalistically followed up at one and two years after the end of the controlled trial stage. The results of the 3 year follow up, show that none of the treatment groups differed on any of the five clinical and functional outcomes (parent and teacher-rated ADHD and oppositional symptoms, reading achievement scores, social skills, and functional impairment). Also, there were no differences in substance use or delinquency with the exception of a slightly lower rate of substance use among those in the psychosocial treatment group. While change had been steepest during the first 14 months (mostly in the methylphenidate group), this leveled off and at three years all groups showed a similar improvement; the methylphenidate group did not deteriorate but the other groups caught up.

These results have caused much speculation but highlight several issues. First, it suggests the Rolls Royce model (medication *plus* psychosocial treatment involving the child, family, and school), is *not* more effective in the long run than any of the other treatments, including the often maligned community care. The combined treatment is seen as the ideal but is rarely delivered in practice due to high cost and burden for parents and schools. Second, there seems to be a 'growing out of' or developmental factor at work. Epidemiological and follow up data have consistently reported a reduction in ADHD symptoms with increasing age—though some individuals continue exhibiting problems. This is also consistent with findings that brains of ADHD children, rather than showing an abnormal development (like in autism), *mature later*. The MTA did not include a placebo group to conclude that regression to the mean or maturation itself are the reasons for improvement. If that were the case, it is possible that helping families and schools contain children's ADHD behaviors during the mid and late primary school years with minimal interventions (e.g., parent management training) may be enough for a proportion—but not all—of these children to get better, or for medication to be required mostly during this developmental period. Finally, the MTA confirmed that methylphenidate can cause growth and weight retardation, particularly during the first year of treatment.

While results of one study rarely justify drastic changes of practice, the findings underscore the complexity of ADHD, show that stimulant drugs are far from being a silver bullet, and that there is much that we don't yet know. This does not mean that stimulants no longer have a place in the treatment of ADHD. However, clinicians should be circumspect when assessing the need for ongoing treatment (e.g., through medication breaks). Much remains to be done to clarify who benefits the most from medication, at what developmental point stimulants are most useful, and for how long medication should be taken. It is also not known whether these results apply to the slow-release formulations (which may enhance adherence) and to atomoxetine (a non-stimulant drug). Parents, often caught in the bind of concerns about their children taking medication and fear of the consequences of non treatment, might be comforted by these findings.

Question 6
Does ADHD exist in adults?



As shown in the figure (unpublished data from the Australian National Survey of Mental Health and Wellbeing), symptoms of ADHD decrease with increasing age. However, ADHD is increasingly diagnosed and treated in adults. Persistent inappropriate features like inattention and impulsivity leading to impaired social or occupational functioning are salient. Some symptoms should have been present during childhood, since adult onset of ADHD is not regarded as valid. This requires corroborative evidence (e.g. from parents, siblings, reports of consultations in childhood, school reports). Assessment is straightforward when a well documented childhood history of ADHD exists.

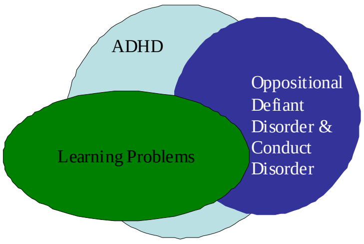
The prevalence of ADHD in adults is unknown but follow-up of ADHD children suggests prevalence under 1% at age 20 and half of that by middle age. Comorbididy, particularly substance abuse, mood and personality disorders, is common and should be considered in treatment.

Adults with clear ADHD benefit from methylphenidate or dexamphetamine at a similar dose to children. Cardiovascular side-effects are monitored, particularly in those with other risk factors such as hypertension. Atomoxetine can be used as an alternative. A second opinion is desirable when diagnostic uncertainty prevails, there is associated substance abuse, or larger than average dosage is required. Prescribing doctors need to seek authority from regulatory authorities since stimulants are addictive drugs (S8).

Question 7

Describe the learning disorders and their association with ADHD

Common problems that often co-occur with ADHD



Specific developmental delays in acquiring language and motor skills or learning abilities (reading, writing — often subsumed under the rubric of 'dyslexia'— and number skills) become apparent in the primary school years. They are often referred to as 'learning disorders' or 'specific developmental disorders'. They are different from developmental disability or mental retardation in that these children have a normal IQ.

Learning disorders are diagnosed only if the child performs markedly below peers (2 or more years behind) *of similar age and intelligence* in tests measuring reading, writing or arithmetic ability (achievement tests). Other mental health problems often coexist, particularly ADHD and behaviour disorders (see Figure).

Specific learning disorders need to be distinguished from school underachievement. Underachieving students have an average IQ but their performance in most subjects is below that expected according to age and intelligence due to factors other than learning disabilities, e.g., school non-attendance, lack of motivation, poor concentration or depression.

Improvement in specific delays can occur with maturation though rarely. Without intervention, secondary emotional and behavioural problems can result from associated social and communication handicaps. Accurate assessment of language, motor and cognitive skills, is essential in planning management. This may encompass speech therapy, motor skills training, modified education with a teacher's aide supporting the child, and alternative methods of learning and communication such as a computer key board instead of handwriting.

Because of the frequent co-occurrence of ADHD and learning problems and because stimulants increase people's ability to concentrate, some practitioners and lay people think that ADHD medication is indicated for the treatment of learning disorders. This is not so; learning problems require specific treatment, mostly based on remedial education principles. Stimulants are indicated only if there is comorbid ADHD.

CASE FOUR

Short case number: 4_24_04
Category: Mental Health and Human Behaviour
Discipline: Psychiatry
Setting: Hospital Emergency
Topic: Alcohol poisoning and conduct disorder

CASE

Oliver Reed, aged 13, was brought in unconscious by mates to the emergency department of your hospital. He smelled strongly of alcohol and had a blood alcohol content of 0.25%. A diagnosis of acute alcohol intoxication is made. When his parents come to hospital, they describe a long history of difficult behaviour, non-compliance, temper outbursts and, more recently, truancy, staying away from home overnight and alcohol use. They despairingly say "We do not know what to do; all his friends seem to be doing the same thing. We are at the end of our tether"

QUESTIONS

1. What would you explain to Oliver's parents about adolescent drinking?
2. Describe the symptoms and management of acute alcohol intoxication
3. What are the key strategies to reduce alcohol consumption in adolescents?
4. On further assessment of Oliver, the mental health team consider whether he may have a conduct disorder. Summarise the epidemiology, symptoms, diagnosis, treatment and outcome of oppositional defiant and conduct disorder.

Suggested reading:

- Management of Mental Disorders (fourth edition) Volume 2. World Health Organisation, Collaborating Centre for Evidence in Mental Health Policy. Sydney 2004. Pp516-540.

Other resources

- Commonwealth of Australia: Teenagers and Alcohol: A Guide for Parents http://www.alcohol.gov.au/internet/alcohol/publishing.nsf/content/brochure-teen-alcohol

- Donaldson L. Guidance on the Consumption of Alcohol by Children and Young People. Department of Health, United kingdom, 2009. http://www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_110258 [need an Australian resource here not UK]

- National Institute for Health and Clinical Excellence. Alcohol-Use Disorders: Preventing the Development of Hazardous and Harmful Drinking. NICE public health guidance 24. National Institute for Health and Clinical Excellence: London, 2010. http://www.nice.org.uk/nicemedia/live/13001/48984/48984.pdf

ANSWERS

Question 1
What would you explain to Oliver's parents about adolescent drinking?

Wide variation exist in young people's drinking and attitudes to alcohol around the world, influenced by family, peers, schools, ethnic and religious upbringing, media, advertising, and national and cultural contexts. Uptake of alcohol consumption in early adolescence can mark both a rite of passage between childhood and adulthood, and also a phase of limit-testing, transgression or deviance from adult social norms. Alcohol use is particularly marked in adolescents with other psychiatric disorders, especially conduct disorder.

Cross-national studies have noted the emergence of heavy episodic consumption or 'binge drinking' for both young women and young men, in higher and lower income groups. Drunkenness has become in some youth cultures the goal of these drinking sessions, rather than a negative side effect of heavy drinking. It has been suggested that the reason for the growth in young people's heavy sessional consumption is the globalisation of cheap and high strength alcoholic beverages, associated with the expansion of an increasingly alcohol-oriented night time economy appealing to youth and young adults. Evidence of convergence in drinking patterns between young women and young men in some developed countries has been linked to young women's growing equality and educational and employment opportunities.

The proportion of Australian secondary school students drinking in the previous week (current drinkers) during the first years of the 21st century increased with age from 10% of 12-year-olds to 49% of 17-year-olds. Prevalence of alcohol use among students aged 12 to 15 decreased between 1999 and 2005: 87% had ever tried alcohol in 1999 compared with 82% in 2005. A similar decrease was found for those drinking during the previous month (43% and 34% respectively). However, this reduction in drinking was not observed among 16 and 17 year olds. The Australian data—broadly comparable to data elsewhere—shows that parents are the most common source of alcohol among students who drank in the past week, with 37% of males and 38% of females indicating their parents gave them their last drink. Approximately 20% of students who were current drinkers had asked someone else to buy alcohol for them and this person was most likely to be a friend aged 18 years (the legal drinking age in Australia) or over. Spirits (e.g. vodka, scotch, rum), in either the premixed or non-premixed form, were the most common types of drinks consumed by current young drinkers.

Alcohol use among adolescents in the US is more widespread than use of any of the illicit drugs, although consuming alcohol is legally prohibited before the age of 21. In 2008, almost three out of every four 12th-grade students (72%) had tried alcohol and over two fifths (43%) were current drinkers—used alcohol in the previous month. Of greater concern is drinking to the point of inebriation; in this line 18% of 8th graders, 37% of 10th graders, and 55% of 12th graders said they had been drunk at least once in their lifetime. The prevalence of drunkenness during the previous month is high: 5%, 14%, and 28%, respectively, for grades 8, 10, and 12. Alcohol use is more common in males than in females. For example among 12th-graders, daily drinking was reported by 4.0% males compared with 1.7% of females. African-American students show the lowest rates of alcohol use.

Almost 90% of 15 year olds in the UK have tried alcohol, although the proportion of those aged 11–15 who drink alcohol has fallen since 2001. However, those who do drink alcohol consume larger amounts and more often. Binge drinking—defined as five or more drinks in a session on at least three occasions in the previous month—appears to be higher among European than US school students. Binge drinking is particularly prevalent in Ireland (32%), the Netherlands (28%), and the UK (27%). Following an increase in the level of binge drinking in UK teenage boys between 1995 and 1999, rates fell slightly by 2003. On the contrary, binge drinking among teenage girls increased between 1995 and 2003 so that by 2003 teenage girls in the UK were more likely than boys to binge drink.

College/university drinking is also a problem. In the US, where most of the available research comes from, about 20% of college students met diagnostic criteria for alcohol abuse or dependence in the preceding 12 months. Alcohol use is the most frequent reason for emergency medical care and cases of campus discipline. Risk factors for college drinking in the US include:

- Living arrangements—rates are higher among students living in fraternities and sororities followed by on-campus housing (e.g., dormitories, residence halls), and living off-site (e.g., in apartments). Students who live with their families drink the least.
- College characteristics— excessive alcohol use is more likely in colleges where fraternities and sororities dominate or where athletic teams are prominent.
- Being a first-year student seems to also increase the risk—many students initiate heavy drinking during the early days of college life (the first 6 weeks seem to be critical). This is paradoxical because during their high school years, those who go on to college tend to drink less than their peers who do not intend to progress to college.
- Other factors, such as pricing and availability of alcohol in the area surrounding a campus, the institutions' alcohol policies and their enforcement.

Alcohol is a causal factor for accidents, injuries and harm to both drinkers and people around them, be they family members, friends, or simply persons who happen to be in their proximity. It is a cause of reduced school, college and job performance, absenteeism, family problems, suicide, homicide, and crime. It is a leading cause of injuries and deaths from motor vehicle accidents. It is also a contributory factor for risky sexual behaviour, sexually transmitted diseases, and HIV infection. Alcohol is a potent teratogen with a range of negative outcomes to the fetus, including low birth weight, cognitive deficiencies, and fetal alcohol disorders. Alcohol is neurotoxic to brain development, which is manifested in mental retardation in children with fetal alcohol syndrome, and it inhibits brain maturation, especially the development of the frontal lobes and consequent adaptive decision-making. Alcohol is also a major cause of acquired brain damage in later years. The harm caused by alcohol is more marked during adolescence due to the brain still developing.

Question 2
Describe the symptoms and management of acute alcohol intoxication

> *On February 19, 1980, Bon Scott, the Western Australian lead singer of one of the most successful rock bands ever, AC/DC, passed out in a car in his way home after a night of heavy drinking. A friend left him to sleep in the car because he could not be awakened. A few hours later Bon Scott was found lifeless and was rushed to London's King's College Hospital where he was pronounced dead on arrival. At the coronial enquiry the cause of death was listed as "acute alcohol poisoning."*

Heavy episodic (binge) drinking in youth is a key contributor to alcohol poisoning. Successful treatment begins with out-of-hospital recognition. Non-clinically trained individuals like parents, teachers, and peers may recognize and associate slurred speech, poor coordination, and acute broad mood-swings with overt manifestations of intoxication. However, without recognition and medical attention, a young person that initially appears mildly intoxicated one moment may easily go unnoticed the next, only to become unresponsive with impending alcohol poisoning and potential death. Outside of the hospital setting, it is difficult to predict which intoxicated youth may go on to alcohol poisoning. The severely intoxicated youth can be found to be unresponsive, cold (hypothermic), with a slow heart rate (bradycardic) and a low blood pressure (hypotensive). Peers, parents, and police need to carefully monitor if the youth becomes unconscious and refrain from giving food or liquids as this may exacerbate nausea and vomiting.

The clinical manifestations in a severely intoxicated youth known to have co-ingested other drugs may vary widely and depend on the pharmacological interaction between alcohol and the other drugs taken. A major life threat is obstruction of the airway due to loss of gag and cough reflex and ensuing aspiration of stomach contents. Alcohol is a potent central nervous system depressant that can lead to cardiopulmonary arrest.

Although young persons may have stopped consuming alcohol some time before their presentation to the emergency department, blood alcohol level may continue rising through the initial medical assessment, resuscitation, and treatment phases.

The mainstay emergency care treatment objective is aggressive respiratory and cardiovascular supportive care. In youth who are intoxicated but awake with a secure airway at the initial presentation, the emergency treatment team should perform a thorough focused physical exam; the objective is to identify evidence of traumatic injury that may be masquerading as 'intoxication' or co-existing with severe intoxication.

Question 3
What are the key strategies to reduce alcohol consumption in adolescents?

The strategies to reduce alcohol consumption among adolescents are very similar to those for adults. A crucial influence on the amount alcohol-related harm in a society and on the proportion of people who drink harmfully is the overall level of alcohol consumption (e.g., *per capita* consumption). A recurring theme in the debate about measures to reduce alcohol-related harm is whether it is better to concentrate on universal interventions or to target subgroups of the population at particularly high risk of harm. The paradox is that the majority of alcohol-related harm in a community occurs not in the heaviest drinkers but in those whose consumption is at lesser levels.

The arguments in favor of universal measures are scientifically compelling. Increasing the price of alcohol trough taxation — particularly minimum pricing (when the minimum price paid for gram of alcohol in beverages is set by legislation) — is the most effective means of reducing alcohol consumption and alcohol related harm. Price rises delay the age when young people start to drink, reduce the number of drinking bouts and binge drinking, reduce the amount of alcohol consumed on each occasion, and slow progression towards drinking larger amounts.

Most countries have passed laws regulating the minimum age at which alcohol can be consumed in licensed premises, the more common being 18 years (21 years in the US). The implementation of these laws leads to clear reductions in drink–driving casualties and other alcohol-related harms. However, the full benefit of a higher drinking age is only realized if the law is enforced. The greater the number of alcohol outlets and the wider the hours of trade the greater the alcohol consumption among young people, with increased levels of assault, homicide, child abuse and neglect, and self-inflicted injury.

Almost all countries outlaw driving a motor vehicle with a blood alcohol concentration (BAC) above specified levels, which vary according to country (usually 0.08% or 0.05%). At all levels of BAC, the risk of being involved in a crash is greater for younger people and can be reduced by enforcement of the law via random breath-testing.

The appropriateness of bans on alcohol advertising has been the subject of public and political debate for years. Based on growing evidence, public health advocates generally argue that advertising increases total alcohol consumption and misuse and that children, at least, should be protected from exposure to advertising and marketing of alcoholic beverages.

Question 4
On further assessment of Oliver, the mental health team consider whether he may have a conduct disorder. Summarise the epidemiology, symptoms, diagnosis, and treatment of oppositional defiant and conduct disorder.

> *…his academic performance declined, he became more sullen and argumentative, was occasionally truant or caught lying about his whereabouts, did not adhere to curfews, and had joined in with other troubled teenagers. His mother was sure that he had been stealing money from her purse, possibly to buy cigarettes. The teachers at school had also become concerned about his increasingly unruly behaviour and advised his parents to seek professional help.*

Young people with the kind of problems described in this vignette represent the largest single group of patients seen in child and adolescent mental health settings; they are usually labelled as suffering from oppositional defiant disorder (ODD) or conduct disorder (CD). The cost of these conditions to the individuals themselves, their families, and society is high. Leaving aside personal and family suffering and the burden to health services, most juvenile delinquent acts are perpetrated by individuals with CD.

Oppositional defiant disorder (ODD)

ODD applies to children who lose their temper persistently, argue with adults, annoy people, blame others, are touchy, and have frequent or intense temper tantrums (typical symptoms are shown in the Table below). This disrupts relationships with family and peers, and makes them 'difficult to manage' at school where they are perceived as wilful and non-compliant. About 3% of children and adolescents suffer from ODD, being more common in boys than in girls.

Oppositionality is interactional by its very nature and does not occur in all situations (e.g. when a child is not frustrated). It can be a response to an over-controlling family, i.e. if parents do not allow children a degree of freedom appropriate to their age. It is necessary to establish whether ODD behaviour occurs more often or with greater intensity than is typical of children of similar age and intelligence (e.g., mild defiance is common in teenagers, temper tantrums in toddlers). Psychosocial impairment resulting from these behaviours is a useful criterion. Conduct disorder and not ODD should be diagnosed if the child exhibits antisocial or markedly aggressive acts (see Table for symptoms of conduct disorder). Children with ADHD often show oppositional behaviour and ODD should then be diagnosed concurrently. Children with ODD seldom carry out antisocial or delinquent acts; indeed, their conduct can be normal in many contexts (e.g., at school or in social settings). On the other hand, when it starts in infancy, persists, and occurs in a dysfunctional family context, ODD often evolves into conduct disorder.

SYMPTOMS	
Oppositional defiant disorder	**Conduct disorder**
• Often loses temper • Often argues with adults • Often actively defies or refuses to comply with adults´ requests or rules • Often deliberately annoys people • Often blames others for his or her mistakes or misbehavior • Is often touchy or easily annoyed by others • Is often angry and resentful • Is often spiteful and vindictive	• Often bullies, threatens or intimidates others • Often initiates physical fights • Has used a weapon that can cause serious physical harm to others • Has been physically cruel to people or animals • Has stolen while confronting a victim (including purse-snatching, extortion, mugging) • Has forced someone into sexual activity • Has deliberately engaged in fire setting with the intention of causing serious damage • Has deliberately destroyed other's property (other than fire setting) • Has broken into someone's house, building or car • Often lies to obtain goods or favors or to avoid obligations • Has stolen items of non-trivial value • Often stays out at night despite parental prohibition • Has run away from home overnight • Often truants from school

Conduct Disorder (CD)

Children with CD are characterized by a persistent pattern of breaking rules or age-appropriate social norms. They display behaviours such as bullying; initiating fights, cruelty to people or animals, stealing, mugging, breaking and entering, and running away from home (see Table). CD is not diagnosed when behaviours are part of schizophrenia, bipolar disorder or autism. Depression, ADHD, substance abuse and learning problems are often found in association with CD and should be diagnosed separately. While a large proportion of delinquents have CD, isolated antisocial acts do not warrant the diagnosis. CD can begin in childhood (before 10) or in adolescence. The distinction has practical implications since earlier onset has a worse prognosis.

About 6% of adolescent boys and 2% of girls have CD, prevalence increasing with social disadvantage. It is of concern that the rates have increased in girls, possibly because social and gender-roles have changed since the 1950s. There is a firm link between CD and adult antisocial personality. Antisocial children tend to become both antisocial teenagers and adults. Further, such adults tend to produce antisocial children. On the positive side, many antisocial children, even the disadvantaged, do not 'graduate' into antisocial adolescents or adults. Less is known about protective factors but 'good enough' mothering, higher intelligence and adequate supervision (e.g., parents know their whereabouts) are among them.

Are these children ill or bad? Are they biologically predisposed or the result of a 'sick' society? Some argue that CD simply medicalises 'badness' or shifts the blame from a disturbed family or community to the child; yet others regard antisocial behaviour as resulting from alienation and disadvantage. A contrary view holds that the CD construct is valid for these reasons: features correlate with one another, suggesting a coherent syndrome rather than an aggregate of assorted types of deviance, CD has a genetic component, and conduct problems are recognised in every society and historical period. Nevertheless, the role of environmental factors like parental neglect, sexual abuse, harsh discipline, inadequate supervision and delinquent peers is central.

Treatment

In practice, children with ODD and CD are treated with a variety of psychological, behavioral, or pharmacological approaches, alone or in combination, targeting the child and/or the family. General principles to keep in mind when treating these disorders are:

- They tend to be chronic conditions, more similar to heart disease than pneumonia, and treatment should be tailored accordingly.
- Most guidelines concur that structured psychosocial interventions (e.g., problem-solving, parent management training and family therapy) should be the first line of treatment for ODD and CD.
- Medication (antipsychotic and mood stabilising drugs) has a minimal role but may be indicated in emergency situations and for comorbid conditions. Clonidine may help in reducing aggressive behaviour in children with ADHD.
- Treatment is more likely to be effective when administered early in the course of the disorder. Typically, maladaptive behaviours are continually reinforced; over time, negative perceptions, emotions, and patterns of relating become deeper and more entrenched. Once CD is established, it becomes more resistant to intervention.
- Treatment should involve the parents. In almost all instances, improving parenting skills and parent-child interactions are core goals.
- Comorbid conditions (e.g. ADHD, depression) ought to be identified and, if appropriate, treated. Parental depression, psychosis, or substance abuse should also be noted and treated.
- It is very useful to ascertain children's and families' strengths and build on them in addition to focusing on their problems.
- Dealing with the stress, anger and hopelessness that many of these families experience and achieving some calm and control is often a necessary initial step.
- The goals of treatment need to be realistic and modified as progress occurs. For example, preventing or minimizing drug use or involvement in delinquent activities in adolescents with CD may be a more appropriate initial step than seeking symptom resolution.
- Because these young people usually show disturbance in a variety of settings (e.g., school, home) and impairment in several aspects of functioning, addressing their multiple needs in the various domains is likely to increase effectiveness (multimodal treatment).

- Association with deviant peers is a well established factor that increases the likelihood of conduct problems, delinquency, and drug use, particularly in adolescence. A goal should be to enhance participation in activities with well-functioning peers.
- In the case of ODD, the main goal of treatment is to increase compliance and reduce conflict. Therefore, the treatment plan should include ways of helping the young person become more cooperative, less argumentative, and be better accepted by peers, often in a family therapy context.
- When using medication, the dosing strategy of "start low, go slow, taper slowly" should be followed, particularly with antipsychotic drugs.
- Caution and careful monitoring should be exercised when prescribing stimulants to adolescents with CD, given the high rate of substance abuse in this population.
- Adherence, side effects and drug interactions should be monitored routinely and systematically. In particular, it is important to ascertain whether patients are concurrently using psychoactive substances or complementary therapies that may interact with prescribed drugs.
- Before switching, augmenting, combining, or discontinuing medications because of a lack of response, it should be ensured that patients have received adequate trials (dose and duration) as well as psychosocial interventions.
- Polypharmacy should be avoided whenever possible.

Emergencies

Presentation of young people with ODD or CD to emergency services is not uncommon. Emergencies usually occur following conflict, legal or disciplinary crises that result in the child losing control and becoming aggressive towards the self, others, or property. Children with ODD or CD who present to emergency services tend to have more severe disorders, more aggression, more comorbid conditions, and fewer family and social supports than those without. Police or mobile psychiatric services involvement is often required. Some of the management principles to be followed in these cases are:

- Crisis intervention strategies should be employed before resorting to the use of medication to control behaviour.
- Physical and mechanical restraints and locked seclusion should be used only as a last resort, when all other approaches have failed.
- The choice to use emergency ("stat" or "p.r.n.") pharmacological management should correspond to the risk for potential injury.
- Emergency staff should be aware of the risks and side effects of acute sedation and follow the appropriate protocols.
- When antipsychotic "p.r.n." or "stat" medications are used several times per day to manage agitation or aggression, clinicians should re-evaluate the diagnosis and the adequacy of behavioural and environmental interventions and then readjust the treatment plan and medication regimen. In most cases, physicians should consider using standing antipsychotic medications rather than frequent "stat" medications.
- Contacting and involving protection services may be necessary in some cases (e.g., the parents refuse to take the child home; there are allegations or suspicion of abuse).